

11

For V_{OH} , $V_{BE} = 0$ when $V_{in} = 0$, with $V_{BE} < V_{BE(on)}$, Q1 is fully off, allowing V_{out} to be pulled up fully to V_{CC} .

$$V_{OH} = V_{CC} = 5V$$

For V_{OL} , R_C contends with Q1. With V_{out} expected to be low, assume saturation.

$$V_{CE} = 0.15V, \underline{V_{out} = 0.15V}$$

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{5 - 0.15}{11k} = 0.441mA$$

$$I_B = \frac{V_{in} - V_B}{R_B} = \frac{5 - 0.75}{5k} = 0.85mA$$

For saturation,

$$I_C < \beta I_B$$

$$0.441mA < 150 \cdot 0.85mA$$

$$0.441mA < 127.5mA \checkmark$$

$$\underline{V_{OL} = 0.15V}$$

V_{IL} is when Q1 turns on,

$$V_B = 0.65V$$

At turn on, $I_B = I_C = 0$, so $\frac{V_{in} - V_B}{R_B} = 0$, $V_{in} = V_B$, $\underline{V_{IL} = 0.65V}$

V_{IH} is at onset of saturation,

$$V_{CE} = V_{CE(sat)} = 0.15V, \text{ and } I_C = \beta I_B$$

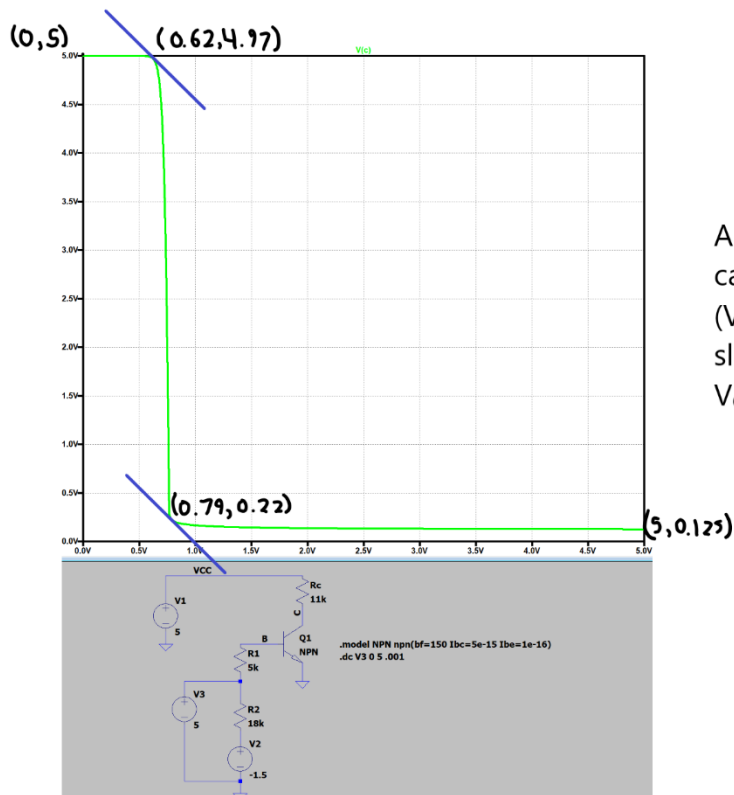
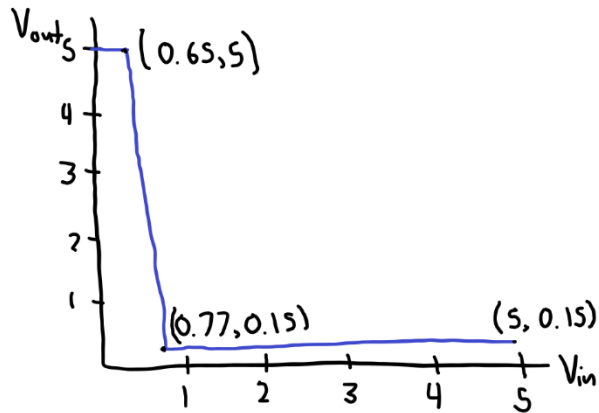
$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{5 - 0.15}{11k} = 0.441mA, \quad 0.441mA = 150 \cdot I_B$$

$$I_B = 2.94\mu A$$

$$\frac{V_{in} - V_B}{R_B} = \frac{V_{in} - 0.75}{5k} = 2.94\mu A, \quad V_{in} = 0.765V$$

$$\underline{V_{IH} = 0.765V}$$

b)



All breakpoints are close to hand calculations. Largest difference is (V_{IH}, V_{out}) , and V_{OL} , where V_{out} at V_{IH} is slightly higher than predicted, and V_{OL} slightly lower.

When $NM_H = 1.75V$, V_{IH} is unchanged, V_{OH} will decline with an output current

$$V'_{OH} - 0.765 = 1.75$$

Assume that all R_C current exits V_{out} . No I_C current

$$I_{IH} = \frac{V_{OH} - V_{BS}}{R'_B} + \frac{V_{OH} - V_B}{R_B}$$

Assume saturation for output low on driven inverter.
 $V_B = V_{BE(sat)}$

$$\underline{I_{IH}} = 0.223 \text{ mA} + 0.353 \text{ mA}$$

$$N = \frac{I_{04}}{I_{14}} = \frac{0.226_m}{0.576_m} = 0.39, \text{ round down, } \underline{0.39}$$

This gate cannot drive itself with $N_{MH} = 1.75$ noise margin.



I_{IH} (when $V_{in} = V_{OH}$) = 571mA, differing by only 5 μ A from hand calculations.